Principles of Atomic Force Microscopy

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Experiments

1.1 Calibration and Characterization of Sensor and Actuator

1.1.1 Offset adjustment and Lorentz Characterization

After adjusting the offset of the deflection readout to obtain a 0 V deflection signal from the strain gauge, we were able to characterize the Lorentz actuator. We changed the supply voltage of the Lorentz actuator from −500 mV to +500 mV and measured the change in cantilever displacement. We measured

$$h_{\text{max}} = 5.58 \text{ cm} \mapsto 500 \text{ mV} \tag{1.1}$$
$$h_{\text{min}} = 5.31 \text{ cm} \mapsto -500 \text{ mV} \tag{1.2}$$

which results in

$$\frac{\Delta V}{\Delta x} = \frac{1000\,\text{mV}}{0.27\,\text{cm}} = 3704\,\frac{\text{mV}}{\text{cm}}. \tag{1.3}$$

The cantilever therefore only moves a tiny bit over its whole voltage range. It is noted though, that in normal use, that is oscillating, this tiny displacement is more than enough to create much higher amplitude deflection at the measuring tip of the cantilever.

1.1.2 Resonance Frequency Characterization

In this step we applied a frequency sweep to the Lorentz actuator to extract its resonance frequency. The resulting measurement is given in figure 1.1. From the plot we can estimate the resonance frequency of the system to be around $f_{\text{res}} \approx 21$ Hz

1.1.3 Force Curve

By applying a triangular voltage to the Lorentz actuator the tip first approaches, touches and then leaves the sample again. This was done, both

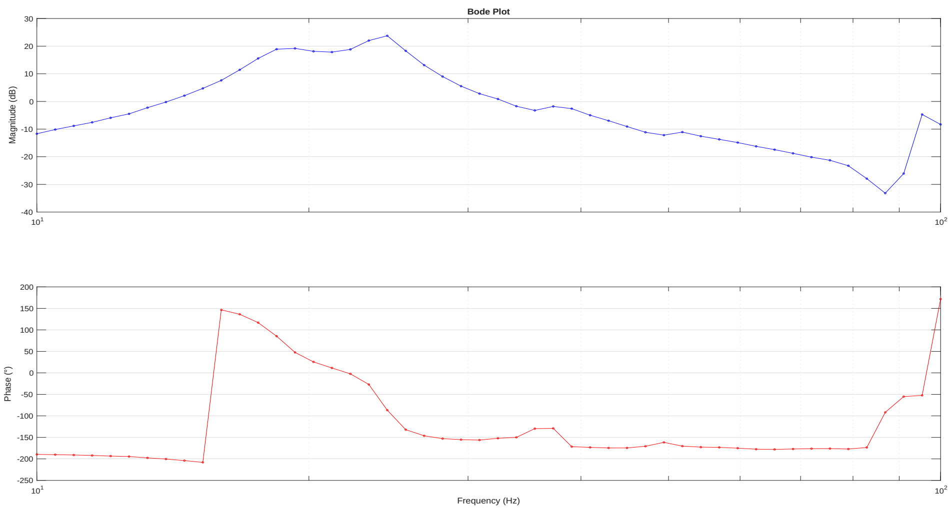

Figure 1.1: Bode plot of the measured frequency response.

with a sticky and non sticky surface. The measured voltage deflection resulting from the bending of the beam upon touching the non sticky surface is given in figure 1.2. In green, we can see the applied voltage, while the measured deflection is plotted in orange. At the top end of the triangle voltage, near its peak, the tip of the cantilever does not yet make contact with the surface. As a result, there is no deflection, and the sensor readout remains flat. Once the tip touches the surface, the cantilever begins to bend, and the deflection signal decreases linearly as the applied voltage on the Lorentz actuator continues to drop. When the voltage rises again, the cantilever gradually straightens, and the deflection signal increases linearly, until the tip loses contact with the surface. At this point, the deflection signal levels off once more, indicating no further interaction. The second test was performed with a sticky surface, resulting in figure 1.3. The resulting measurement is quite similar, with one big difference. Upon retraction, we can see the cantilever first sticks to the surface, and then abruptly snaps off. This can be directly attributed to the stickiness of the surface and is a perfect example of how the atomic force microscope functions. We can clearly make out the surface structure just from the deflection readout of our cantilever.

1.2 Contact Mode Operation

In this mode, the cantilever is brought into contact with the surface, and a control loop is used to maintain constant deflection. Variations in surface height are directly compensated by the controller, which adjusts the Lorentz

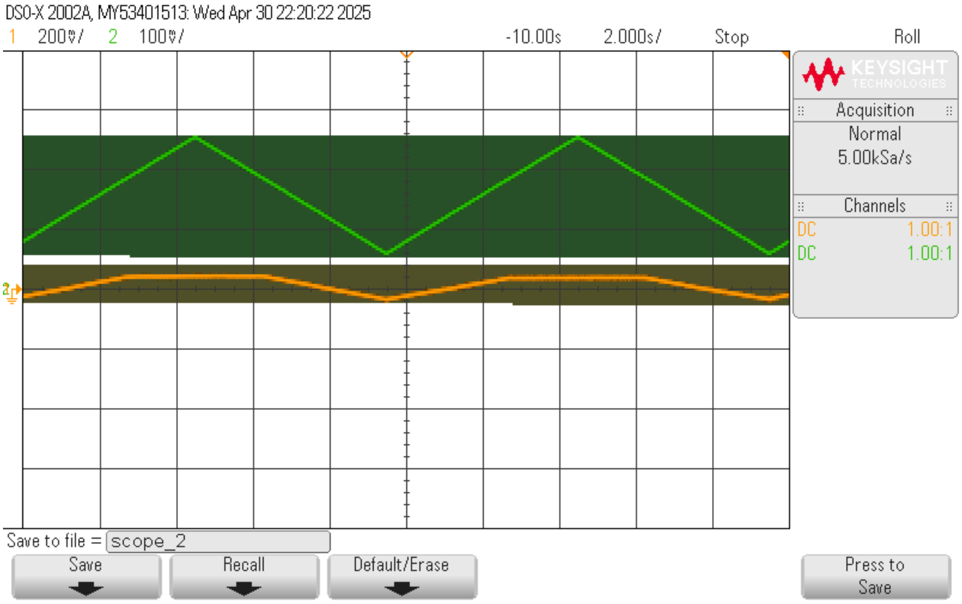

Figure 1.2: Scope Screenshot for the non-sticky surface. In green the applied voltage and in orange the measured deflection of the cantelever.

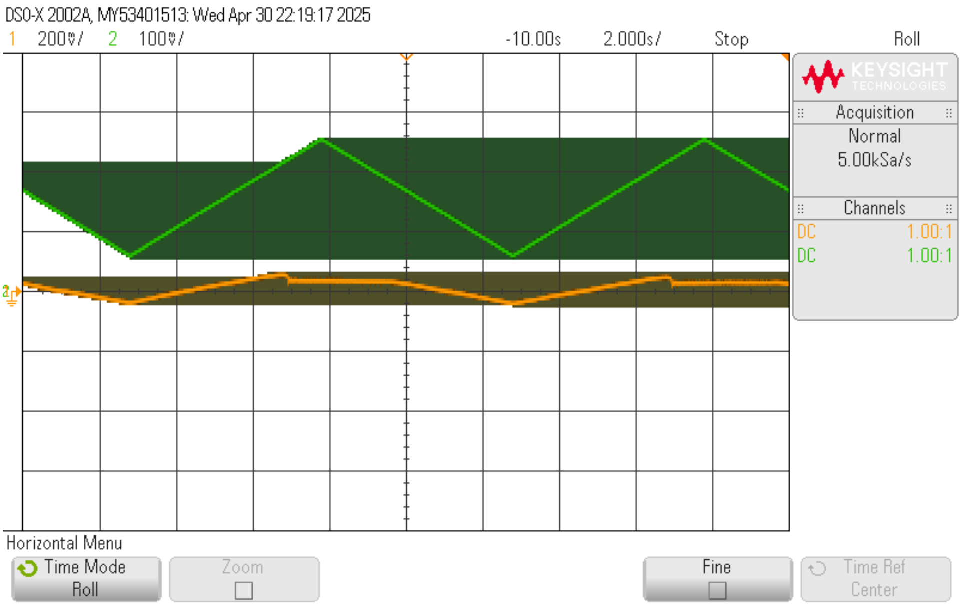

Figure 1.3: Scope Screenshot for the sticky surface. In green the applied voltage and in orange the measured deflection of the cantelever.

actuator to keep the cantilever bend unchanged. As a result, the controller output can be used as a direct measurement of the surface topography.

1.2.1 Controller Implementation

Using the pure integral controller circuit from Fig. 3 in the lab-script we can estimate the bandwidth

$$f_{\mathrm{BW}} = \frac{1}{2\pi RC} = \frac{1}{2\pi \cdot 110 \times 10^3\ \Omega \cdot 100 \times 10^{-9}\mathrm{F}} \approx 14.5\ \mathrm{kHz} \tag{1.4}$$

of the pure integral controller.
We adjusted the z-height of the sample such that significant deflection is detected by the readout electronics. Then we applied a constant voltage at the non-inverting input, while monitoring the integral controllers output. The voltage was chosen such, that the controller output was about 0 at the current z-height we chose. Then the controller output was connected to the Lorentz actuator, closing the feedback loop. This setup procedure effectively calibrates the controller to the chosen z-height as ground level. If the sample surface rises and pushes the cantilever upward, the feedback controller reduces the actuator force to ease the pressure and maintain constant deflection. Conversely, if the surface drops, the controller increases the actuator force to ensure the cantilever continues to apply the same contact force to the sample.

1.2.2 Z-height characterization

With the closed feedback controller working, we can now measure the sample topology via the controller output. Pushing the sample along the calibration track with increasing height, figure 1.4 was recorded. Ideally, we expect the deflection signal (orange) to peak every time the height of the sample increases. Then the controller output (green) should increase to bring the deflection signal back down to the same level. This only worked partially. We can clearly see the controller output increase after the deflection signal dips. However, the controller seems to overcorrect, with the deflection signal returning to a slightly higher level than before the height jump on the sample. Per topography increase, our controller output jumped about

$$V_{\mathrm{jump}} = 170\ \mathrm{mV} \tag{1.5}$$

with corresponds to a height increase of

$$Z_{\mathrm{jump}} = 250\ \mu\mathrm{m}, \tag{1.6}$$

which results in

$$S_{\mathrm{z}} = \frac{250\ \mu\mathrm{m}}{170\ \mathrm{mV}} = 1.47\ \frac{\mu m}{\mathrm{mV}} \tag{1.7}$$

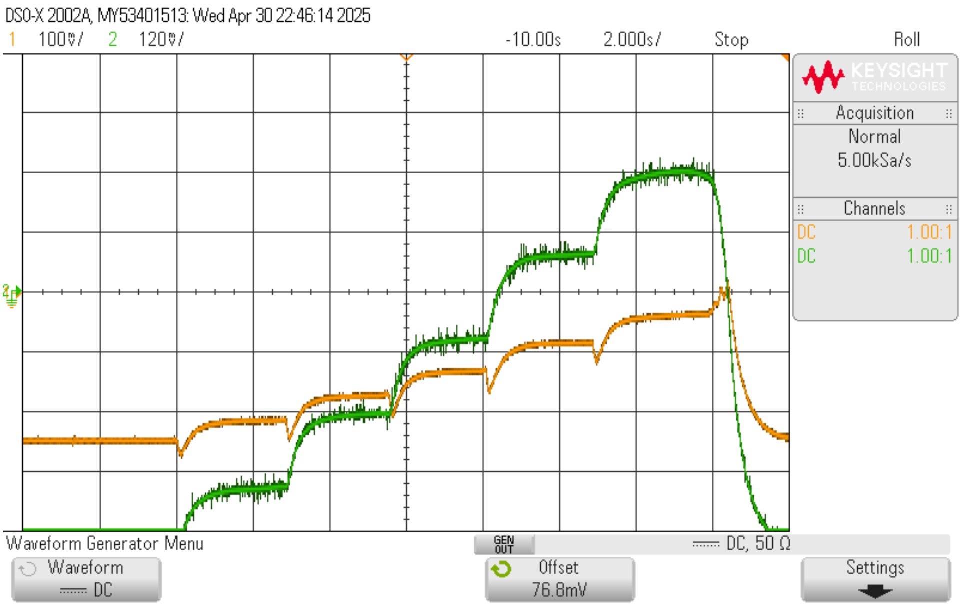

Figure 1.4: Scope Screenshot captured while moving the sample underneath the cantilever. Green: The controller output. Orange: the deflection readout

1.2.3 Closed Loop System Identification

In this section, we measured a closed loop frequency response. The results can be seen in figure 1.5. The magnitude response is qualitatively reasonably similar to the open loop response seen in figure 1.1 with a peak at around 21 Hz after which the magnitude response rolls off.

1.3 Tapping Mode Operation

In this mode, the cantilever continues to tap up and down. We approached the sample in such a way that the deflection signal reduced about 30 %.

1.3.1 Sticky vs. Non-Sticky Surface

We approached both the sticky and non-sticky sample as described above. From this, we obtained two oscilloscope screenshots. Comparing the measurements from the non-sticky surface in figure 1.6 and the sticky surface in figure 1.7,unfortunately, we can barely see any difference. We would expect the sticky surface to decrease the amplitude and also increase a phase delay. Investigating the sample, we found that the sticky sample is barely sticky anymore, which explains our very similar measurements.

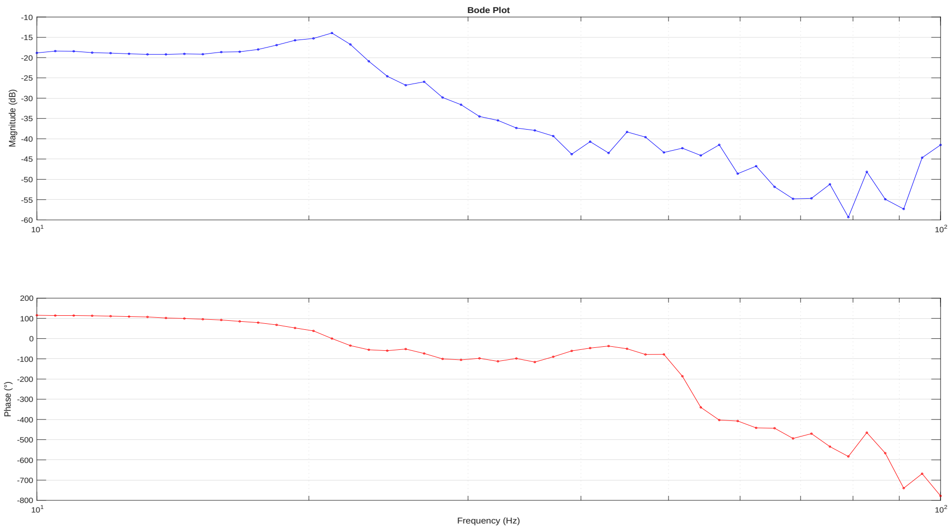

Figure 1.5: Closed loop frequency response.

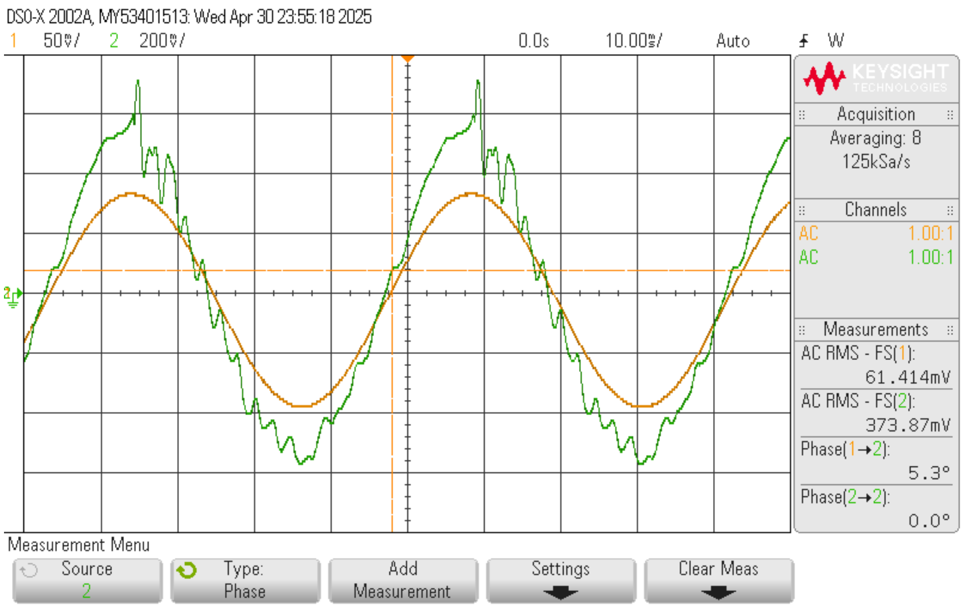

Figure 1.6: Tapping mode non-sticky sample. Orange: Actuation signal. Green: Deflection signal

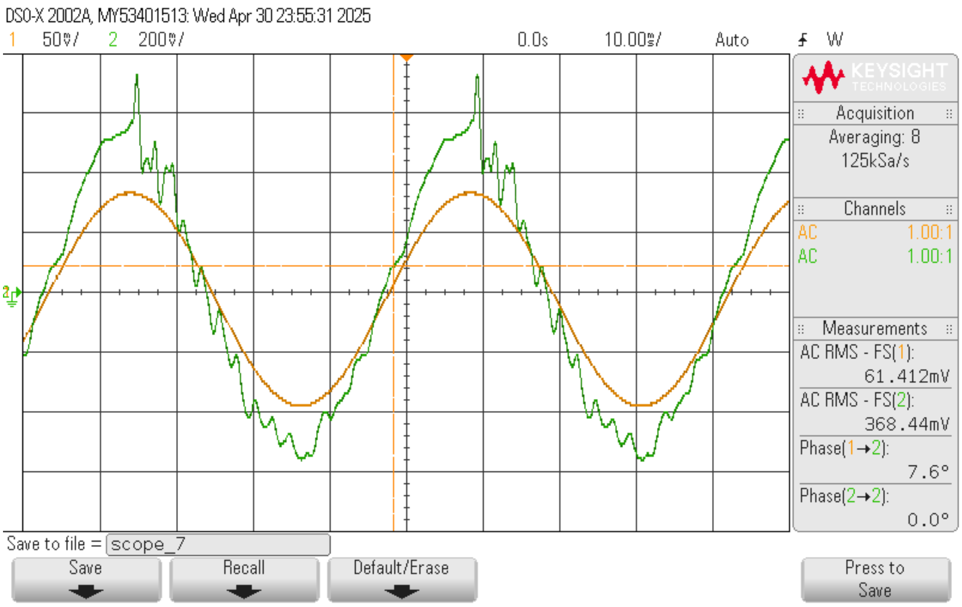

Figure 1.7: Tapping mode sticky sample. Orange: Actuation signal. Green: Deflection signal

1.3.2 Direct Demodulation

To measure the topography of the sample via the controller output like in the contact mode operation, we first need to demodulate the deflection signal. The continuous oscillation of the cantilever is reflected in the strain gauge measurement, which becomes

$$\Delta R = AK \sin(\omega t + \phi_c), \tag{1.8}$$

with the oscillation amplitude A, the oscillation phase ϕ and the strain gauge sensitivity K. By applying a voltage with the same frequency

$$U_s = U_0 \sin(\omega t + \phi_0) \tag{1.9}$$

directly on the measurement bridge circuit, the combined measurement becomes

$$\begin{aligned} U_{\text{out}} &= AK \sin(\omega t + \phi_c) \cdot U_0 \sin(\omega t + \phi_0) && (1.10) \\ &= \frac{U_0 AK}{2R}[\cos(\phi_0 - \phi_c) - \cos(2\omega t + \phi_0 + \phi_c)] && (1.11) \end{aligned}$$

the second term in (1.11) can be filtered out by a low pass filter, leaving us with

$$U_{\text{out}} = \frac{U_0 AK}{2R}[\cos(\phi_0 - \phi_c)]. \tag{1.12}$$

Now aligning

$$\phi_0 = \phi_c, \tag{1.13}$$

by changing ϕ_0, we get

$$U_{\text{out}} = \frac{U_0 AK}{2R}. \tag{1.14}$$

This result is especially nice, because we can now simply use our controller from the contact mode operation to measure the topography.

1.3.3 Tapping Mode Sample Measurement

Connecting everything correctly and making sure our demodulation works correctly, we were able to measure the test sample. The results of which can be seen in figure 1.8. The controller output is green again, while the deflection readout is orange. The feedback control does generally work, increasing its output to compensate for each dip in the deflection measurement. However, we do see the same issue as in contact mode operation. The controller does overcompensate for each dip, with the deflection error settling at a higher deflection as before the respective dip. It is also noted, that the control loop takes significantly longer to correct for any change in topography. This can mainly be contributed to the introduced low pass filter which suppresses any rapid changes.
For the same 250 μm jump in height, the controller output in tapping mode jumps about 110 mV. Resulting in a sensitivity of

$$S_{\text{z}} = \frac{250\ \mu\text{m}}{110\ \text{mV}} = 2.27\ \frac{\mu m}{\text{mV}}, \tag{1.15}$$

which is higher, but still reasonably close, to the in (1.7) calculated sensitivity in contact mode operation.

1.4 Report Questions

1.4.1 Height Map

We unfortunately only measured the calibration track and can therefore only convert this into a height map.

Contact Mode Measurement

with the calculated sensitivity

$$S_{\text{z}} = 1.47\ \frac{\mu m}{\text{mV}} \tag{1.16}$$

and a jump voltage difference of

$$V_{\text{jump}} = 170\ \text{mV} \tag{1.17}$$

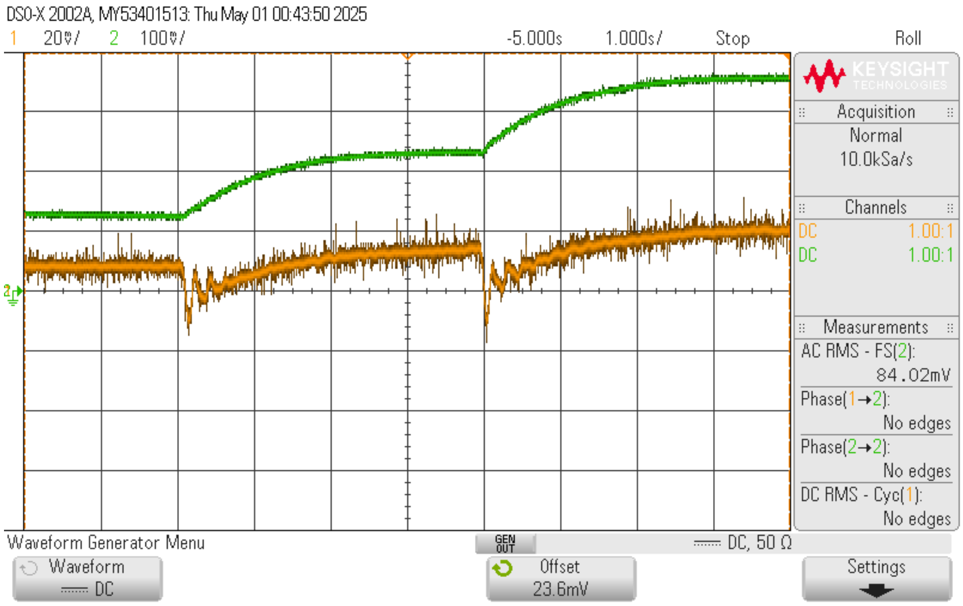

Figure 1.8: Tapping mode, calibration step measurement. Green: Controller output. Orange: Deflection signal

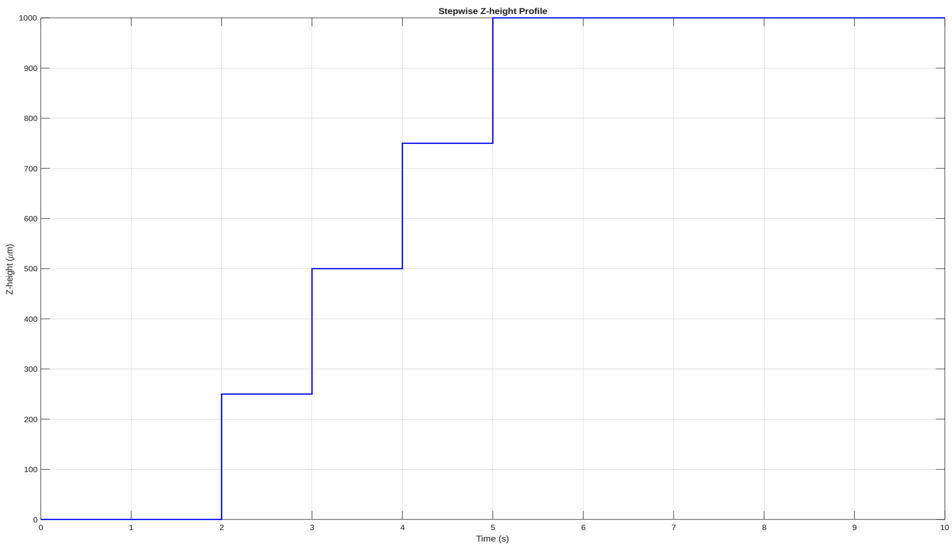

Figure 1.9: Height map of the calibration sample.

we can create a height map 1.9. The result of this is of course not very surprising as this is the sample we used to calibrate the measurement. However it still demonstrates the core idea. With (1.16) we could measure the height of any given sample.

Tip Influence

The size and form of the tip used in the AFM measurement directly influences the achievable resolution. A wide tip might not fit in small grooves on the sample. Similar to a big tire, rolling over a small whole in the road. This of course, distorts the measurement. Therefore the tip size corresponds to the lateral resolution. Meaning how close can two neighboring features on the sample be, such that the AFM still truthfully captures them.
The vertical resolution on the other hand is governed more by the feedback control bandwith and mechanical vibrations.

1.4.2 Closed Loop Control Bandwidth

The control bandwidth is defined as the frequency at which the magnitude response drops below 3dB. Looking at figure (1.5), the magnitude is always way below zero, so we defined the bandwith as the point at which the magnitude drops 3 dB from the peak of the frequency response. This yields a bandwith of about

$$f_{\mathrm{BW}} = 22\ \mathrm{Hz}. \tag{1.18}$$

1.4.3 Tip-Sample Interaction

Contact Mode

In contact mode, the tip is always in contact with the sample. This has both advantages and disadvantages. As seen in the comparison of figure 1.4 and figure 1.8, the contact mode feedback control is much faster. This also results in a faster overal measurement time. The importance of speed, of course, depends on the context of the measurement. Also depending on the sample, the constant contact might also be an issue. The sample is usually moved underneath the measuring tip during the measurement. For fragile sample surfaces the tip might change or even destroy part of the sample, which invalidates any measurement results.

Tapping Mode

The tapping mode basically inverts the up- and downsides of the contact mode operation. As already discussed, measuring-times using tapping mode, might be significantly slower. On the other hand, tapping mode is much gentler towards the sample surface, as no significant lateral force acts on the sample. For softer sample surfaces tapping mode might be the only option.